

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Forecasting Pit stop decisions for Formula 1 race strategy optimization.
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Chosen major:	M.Sc. Computing (Data Analytics)
Supervisor	Dr. Michael Scriney
Date of Submission	21-11-2025

Section 2: About your Practicum

What is the topic of your proposed practicum?

"Forecasting Pit Stop Decisions for Formula 1 Race Strategy optimization" is the topic of our proposed practicum. This practicum examines how pit stop decisions can be forecasted using open datasets, which consist of lap times, pit events and uncertain race events. It compares different modelling approaches to assess how accurately they can predict optimal pit timing and evaluates which features most strongly influence these forecasts. By analysing model behaviour and hyperparameter choices, the project aims to identify the most effective predictive setup. The findings are intended to support more informed and data-driven strategy decisions in Formula 1 racing.

Please provide details of the papers you have read on this topic

[1] Carrasco Heine, Oscar Felipe and Thraves, Charles, On the Optimization of Pit-Stop Strategies via Dynamic Programming (April 29, 2021). Available at SSRN: <https://ssrn.com/abstract=3769652> or <http://dx.doi.org/10.2139/ssrn.3769652>

[2] Heine, O.F.C., Thraves, C. On the optimization of pit stop strategies via dynamic programming. *Cent Eur J Oper Res* **31**, 239–268 (2023). <https://doi.org/10.1007/s10100-022-00806-4>

[3] A. Greasley, G. Panchal and A. Samvedi, "The Use of Simulation with Machine Learning and Optimization for a Digital Twin-A Case on Formula 1 DSS," 2022 Winter Simulation Conference (WSC), Singapore, 2022, pp. 2198-2209, doi: 10.1109/WSC57314.2022.10015299. <https://ieeexplore.ieee.org/document/10015299>

[4] L. Paparusso, M. Riani, F. Ruggeri and F. Braghin, "Competitors-Aware Stochastic Lap Strategy Optimisation for Race Hybrid Vehicles," in *IEEE Transactions on Vehicular Technology*, vol. 72, no. 3, pp. 3074-3089, March 2023, doi: 10.1109/TVT.2022.3215171

https://ieeexplore.ieee.org/abstract/document/9921276?casa_token=vLvSycbrKFgAAAAA:fnZw8mVGLohrKmtNkET18AJyXan-JDPmcTfGZHE0VLI_KjJ4mfWIVycl43nmz38fnr_22-kPCk8

[5] P. Duhr, D. Buccheri, C. Balerna, A. Cerofolini and C. H. Onder, "Minimum-Race-Time Energy Allocation Strategies for the Hybrid-Electric Formula 1 Power Unit," in *IEEE Transactions on Vehicular Technology*, vol. 72, no. 6, pp. 7035-7050, June 2023, doi: 10.1109/TVT.2023.3237388

<https://ieeexplore.ieee.org/document/10018239>

How does your proposal relate to existing work on this topic described in these papers?

- The existing technology used in the above research papers is mainly focused on Dynamic Programming, and we plan to bridge the research gap by using machine learning models in the practicum.
- In simple terms, Dynamic Programming is like a driver doing a perfect lap, given the tyre wear percentage and its position, when given a proper race strategy, whereas Machine Learning takes intakes from historical races and learns from the experiences, understands the patterns, and makes strategies based on the current conditions.
- ML provides us with an option to take uncertainty into consideration, which dynamic programming lacks. This model can help us to come up with a better pit stop strategy, if any uncertain race event happens, like a safety car or a strategic move by the opponent. The existing system contains the policies and rules that are made by humans, and it is not flexible. Whereas ML models can learn and make policies by learning from historical data and can adapt to these changes.

What are the research questions that you will attempt to answer?

1. Can ML approaches effectively predict and optimise lap timing in F1 ?
 2. Which features have a major impact on the performance of the model?
 3. To what extent can XAI aid in the communication of pit stop optimisation in F1?
- We will investigate these questions by training multiple ML models with different hyperparameter configurations and thus, forecasting pit stop predictions, which may lead to strategic and optimized decision-making in the Formula 1 races.

- What software and programming environment will you use?

1. The project will be developed within a Python-based data science environment that supports the creation, training, and simulation of analytical models.
2. Frameworks such as PyTorch or TensorFlow, together with Pyro or TensorFlow Probability, will be used to build neural and probabilistic models capable of representing uncertainty.
3. Graph Neural Networks will be designed using libraries like NetworkX and PyTorch Geometric to capture the relationships and interactions between vehicles.
4. Data preparation, feature engineering, and model evaluation will be carried out using NumPy, pandas, and scikit-learn.
5. Visual analysis and interpretation will be supported through Matplotlib, Seaborn, and Plotly, with the option to create interactive dashboards in Power BI or Streamlit.
6. Structured racing and simulation data will be organised and maintained using MySQL Workbench to ensure efficient access and management.

- What coding/development will you do?

1. Sequence-based models such as RNNs, LSTMs or temporal graph neural networks will be used to capture how race information evolves across laps.
2. A conditional lap-time generator, using methods like mixture density networks or normalizing flows, will model the natural randomness in lap-time outcomes.
3. A graph neural network interaction module will represent relationships between cars, including overtakes and undercuts.
4. All components will run inside a simulation framework that updates race states using Monte Carlo sampling for uncertainty estimation.
5. Evaluation and visualisation tools will compare model outputs with real race data and provide an interactive dashboard for scenario testing and what-if analyses.
6. Together, these modules support the project goal of forecasting pit-stop decisions for Formula One strategy optimisation.

- What data will be used for your investigations?

1. The project will utilise multi-source datasets, integrating information from the Ergast API for race and pit data, Kaggle F1 repositories for lap times and telemetry, and the FastF1 library for detailed car state and tyre metrics.
2. A custom integration pipeline will consolidate these sources into a unified MySQL or Parquet-based data structure to ensure consistency and accessibility.

3. Where data gaps exist, such as tyre degradation or fuel proxies, they will be synthetically modelled or approximated using correlated telemetry variables to maintain dataset completeness.

- Is this data currently available? if not, where will it come from?

Yes, the data is available, and here are the links for the datasets :

1. Fast F1 API
<https://github.com/theOehrly/Fast-F1>
2. Formula 1 Race Data
<https://www.kaggle.com/datasets/cjgdev/formula-1-race-data-19502017>

- What experiments do you expect to run?

1. Linear Regression / Ridge: baseline for Δ Position or lap_time_gain. (fast, interpretable).
2. Random Forest: non-linear baseline, robust to missing features.
3. XGBoost / LightGBM : primary supervised model for tabular prediction (Δ Position, Δ Time, Pit_Time).

- What output do you expect to gather?

1. We intend to compile a detailed table documenting the lap times recorded at each pit stop, along with the overall completion time for each driver.
2. This dataset will provide a clear and organized foundation for examining performance trends, comparing outcomes across participants, and supporting the broader analytical aims of the study.

- How will the results be evaluated?

1. The results will be examined in terms of how accurately the model predicts race behaviour and how well it captures the underlying factors that influence strategic decisions during an event.
2. This assessment will involve comparing the model's estimates of lap times, time gaps, and finishing positions with the actual figures recorded in the races used for testing.
3. Statistical measures such as RMSE, MAE, and likelihood-based evaluations will help determine the reliability of the model's forecasts and the extent to which it can

support informed decision-making when optimising pit-stop strategies in a Formula One setting.